

Satvik Tajane

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EDUCATION

Master of Science in Electrical and Computer Engineering

Expected: June 2027

Specialization: Hardware and Software for Machine Intelligence

Northeastern University, Boston, MA

Relevant Courses: Machine Learning, Mobile Robotics, Robot Sensing and Navigation

Bachelor of Technology in Electrical and Computer Engineering

August 2025

MIT World Peace University, Pune, India

Relevant Courses: Control Systems, Robotics, Artificial Intelligence and Machine Learning

TECHNICAL SKILLS

Programming: Python, C++
Development Tools: Gazebo, RViz, Arduino IDE, MATLAB, ROS2(humble,jazzy)
Perception: AMCL, Kalman filters, Particle filter
Path Planning Algorithms: A*, RRT, Dijkstra
Libraries: OpenCV, Pytorch

PROJECTS

Ro-Brew-To: Autonomous Coffee-Making Robot

Expected: July 2026

- Designing and simulating an autonomous pick and place pipeline for a UR3e 6 DOF manipulator with Robotiq 2F-85 gripper to operate a pod espresso machine, integrating perception, motion planning, and orchestration on ROS 2 Jazzy
- Building a custom Gazebo Harmonic world with articulated coffee machine model , overhead RGBD camera sensor
- Implementing marker based 6D pose estimation using OpenCV ArUco detection, projecting pixel coordinates to world frame via camera intrinsics and TF2 transforms for real time object localization
- Developing pick and place motion sequences using MoveIt 2, including Cartesian arc trajectories for lever actuation and constrained planning for pod insertion, targeting full closed loop autonomy from detection to brew completion

Degradation Aware Particle Filter for Robot Localization

January 2026 – April 2026

- Built a custom Gazebo simulation environment integrating LiDAR, camera, and IMU sensor plugins for a ground robot, enabling controlled multi sensor degradation experiments
- Designed a degradation aware particle filter with sensor fallback, introducing 4 levels of synthetic degradation to LiDAR and camera data and using a confidence monitor to dynamically weight sensor contributions
- Outperformed Nav2 AMCL baseline by 1 m in Absolute Trajectory Error (ATE RMSE), demonstrating robust localization under sensor failure conditions

Control Barrier Function for Safe Robot Navigation

September 2025 – December 2025

- Worked on a mobile robotics project focused on safe navigation in warehouse environments using Control Barrier Functions (CBFs)
- Implemented and simulated CBF based safety constraints in Gazebo, ensuring collision avoidance while minimally modifying nominal controllers
- Validated the approach through multi robot simulation scenarios, demonstrating safe and stable navigation in cluttered warehouse layouts

PROFESSIONAL EXPERIENCE

Test and Validation Intern

January 2025 – June 2025

Alpha-ICT, Pune, India

- Engineered and executed automated testing and validation protocols for sophisticated twin HVAC (Heating, Ventilation, and Air Conditioning) systems
- Implemented Robot Framework in Python to achieve precise and repeatable hardware testing results
- Innovated a solution to bypass the flame sensor safety interlock on the HVAC heating unit, enabling continuous, unattended overnight testing and significantly accelerating testing time